

```
topt shutdown);
allow aide_t.newrole.t : fd use ;
```

The *research* command allows us to query a policy for *Type Enforcement* rules. Let us explore the first of these rules—the Allow Rule.

- **Allow Rules:** These specifically allow 'access' to an 'object' by a 'subject'.

Here, access is defined by:

- access permissions—such as read, write, execute, etc

object is defined by:

- the security context called the *target context (tcontext)*
- the class of the object called the *target class (tclass)*. Examples of the target class can be file, dir, socket, etc

subject is defined by:

- the security context called the *source context (scontext)*

A typical allow rule can be described as follows:

Allow the Web Process (Apache server) to read the file (*/var/www/html/index.html*)

If the above rule is not present in the policy, the Apache process will not be able to read a file in its default 'documentroot' folder and will be denied access.

To implement the above allow rule, we need to evaluate *Access Permissions Required*, *Target Context (tcontext)*, *Target Class (tclass)*, and *Source Context (scontext)*

For our example, the results will be as follows:

- Access Permissions Required: *read*
- Target Context (tcontext): (security context of */var/www/html/index.html*)
(*ls -Z /var/www/html/index.html*)
=> system_u:object_r:httpd_sys_content_t:s0
- Target Class (tclass): *file*
- Source Context (scontext): (security context of the *httpd* process)
(*ps axZ | grep httpd*) *=> user_u:system_r:httpd_t:s0*

Taking the above into consideration, our allow rule changes from:

Allow the Web Process (apache server) to read the file (*/var/www/html/index.html*)

TO

Allow the Source Context – *user_u:system_r:httpd_t:s0* permission to read on the class file bearing a Target Context of *system_u:object_r:httpd_sys_content_t:s0*

To search for all allow rules, specify as follows:

```
[root@vbg ~]# research --allow
```

To search for an allow rule that specifically contains *scontext*, *tcontext* and *tclass*, specify:

```
[root@vbg ~]# research -s sccontext -t tcontext -c tclass --allow
```

Since this rule exists in the default targeted policy, let us search for it using the *research* command:

```
[root@vbg ~]# research -s httpd_t -t httpd_sys_content_t -c file --allow
Found 2 av rules:
allow httpd_t:httpd_sys_content_t:file {sock} read getattr lock ;
allow httpd_t:httpd_sys_content_t:file {sock} read getattr lock ;
```

Let us examine the syntax of the allow rule itself. The first word in a rule phrase specifies the type of the rule. Therefore, allow rules in a policy appear as: *allow sccontext tcontext: tclass permissions*

By default, the rules in the targeted security policy do not allow the Web server to read a file of type *tmp_t*.

You can test this by changing the type of the *index.html* file.

```
[root@vbg ~]# chcon -t tmp_t /var/www/html/index.html
```

Now try to open this Web page. You will receive a *forbidden/access denied* error. This is because of SELinux MAC rules. There is no 'allow rule' in the policy to allow this access.

If you want to allow the Apache server to be able to read this file, you will need to insert an allow rule to the

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